

OSLAB PRACTICAL-7 PROGRAMS & OUTPUT

Date: 05-08-26

Program-1 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_7$ cat program1.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global = 10;
static int static_global = 20;

void code_function() {
    printf("Code address:    %p\n", (void *)code_function);
}

int main() {
    int stack_var = 30;
    static int static_local = 40;
    int *heap_var = malloc(sizeof(int));

    *heap_var = 50;

    printf("Process ID:      %d\n", getpid());
    code_function();

    printf("Global address: %p\n", (void *)&global);
    printf("Static address: %p\n", (void *)&static_global);
    printf("Static local:    %p\n", (void *)&static_local);
    printf("Heap address:    %p\n", (void *)heap_var);
    printf("Stack address:   %p\n", (void *)&stack_var);

    printf("\nPress Enter to exit...\n");
    getchar();

    free(heap_var);
    return 0;
}

sainath@Sainath:~/OSLAB/practical_7$ ./program1
Process ID:      647
Code address:    0x58aeb61bc209
Global address:  0x58aeb61bf010
Static address:  0x58aeb61bf014
Static local:    0x58aeb61bf018
Heap address:    0x58aecad582a0
Stack address:   0x7ffcb47a78ec

Press Enter to exit...
```

Program-2 & OUTPUT:

```
sainath@Sainath:~/OSLAB/practical_7$ cat program2.c

#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global = 10;
static int static_var = 20;

void code_function() {
    printf("Code address:  %p\n", (void *)code_function);
}

int main() {
    int stack_var = 30;
    int *heap_var = malloc(sizeof(int));
    *heap_var = 40;

    printf("PID:           %d\n", getpid());
    code_function();

    printf("Global address: %p\n", (void *)&global);
    printf("Static address: %p\n", (void *)&static_var);
    printf("Heap address:  %p\n", (void *)heap_var);
    printf("Stack address:  %p\n", (void *)&stack_var);

    printf("\nVirtual memory mappings:\n");

    char path[50];
    sprintf(path, "/proc/%d/maps", getpid());

    FILE *file = fopen(path, "r");

    if (file == NULL) {
        perror("fopen");
        return 1;
    }

    char line[256];

    while (fgets(line, sizeof(line), file)) {
        printf("%s", line);
    }

    fclose(file);
    free(heap_var);

    return 0;
}
```

```

sainath@sainath:~/OSLAB/practical_7$ ./program2
PID: 649
Code address: 0x643ea495c289
Global address: 0x643ea495f010
Static address: 0x643ea495f014
Heap address: 0x643ec52f12a0
Stack address: 0x7ffc8eef72ac

Virtual memory mappings:
643ea495b000-643ea495c000 r--p 00000000 08:30 11870 /home/sainath/OSLAB/practical_7/program2
643ea495c000-643ea495d000 r-xp 00001000 08:30 11870 /home/sainath/OSLAB/practical_7/program2
643ea495d000-643ea495e000 r--p 00002000 08:30 11870 /home/sainath/OSLAB/practical_7/program2
643ea495e000-643ea495f000 r--p 00002000 08:30 11870 /home/sainath/OSLAB/practical_7/program2
643ea495f000-643ea4960000 rw-p 00003000 08:30 11870 /home/sainath/OSLAB/practical_7/program2
643ec52f1000-643ec5312000 rw-p 00000000 00:00 0 [heap]
7a083e000000-7a083e028000 r--p 00000000 08:30 49760 /usr/lib/x86_64-linux-gnu/libc.so.6
7a083e028000-7a083e1b0000 r-xp 00028000 08:30 49760 /usr/lib/x86_64-linux-gnu/libc.so.6
7a083e1b0000-7a083e1ff000 r--p 001b0000 08:30 49760 /usr/lib/x86_64-linux-gnu/libc.so.6
7a083e1ff000-7a083e203000 r--p 001fe000 08:30 49760 /usr/lib/x86_64-linux-gnu/libc.so.6
7a083e203000-7a083e205000 rw-p 00202000 08:30 49760 /usr/lib/x86_64-linux-gnu/libc.so.6
7a083e205000-7a083e212000 rw-p 00000000 00:00 0
7a083e307000-7a083e30a000 rw-p 00000000 00:00 0
7a083e311000-7a083e313000 rw-p 00000000 00:00 0
7a083e313000-7a083e317000 r--p 00000000 00:00 0 [vvar]
7a083e317000-7a083e319000 r--p 00000000 00:00 0 [vvar_vclock]
7a083e319000-7a083e31b000 r-xp 00000000 00:00 0 [vdso]
7a083e31b000-7a083e31c000 r--p 00000000 08:30 49757 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a083e31c000-7a083e347000 r-xp 00001000 08:30 49757 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a083e347000-7a083e351000 r--p 0002c000 08:30 49757 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a083e351000-7a083e353000 r--p 00036000 08:30 49757 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7a083e353000-7a083e355000 rw-p 00038000 08:30 49757 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
7ffc8eed9000-7ffc8eefa000 rw-p 00000000 00:00 0 [stack]

```

history:

```

420 cd
421 cd OSLAB
422 ls
423 mkdir practical_7
424 cd practical_7
425 gedit program1.c
426 gcc program1.c -o program1
427 ./program1
428 gedit program2.c
429 gcc program2.c -o program2
430 ./program2
431 cat program1.c
432 ./program1
433 cat program2.c
434 ./program2
435 history

```